
IQRA UNIVERSITY

Department of Computer Science

INFORMATION SECURITY

FINAL PROJECT REPORT

Enterprise Network Security Design Using Cisco ASA Firewall with NAT and ACL Policies

Simulated in Cisco Packet Tracer

Submitted By BS(CS) Students

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1. Introduction

This project presents the design and implementation of a secure enterprise network using a Cisco Adaptive Security Appliance (ASA) firewall. The project was carried out as part of the Computer Networks course and simulates a realistic enterprise environment where an internal (inside) network is protected from an external (outside) network through firewall policies, Network Address Translation (NAT), and Access Control Lists (ACLs).

The main purpose of this project is to understand how enterprise firewalls control and secure traffic between an internal company network and the outside internet or a service provider. This includes translating private IP addresses into public IP addresses, allowing only specific traffic in and out of the network, and enforcing security rules for individual users.

1.1 Problem Statement

The goal of this project is to design and manage the security policies of an enterprise network by configuring a network firewall (Cisco ASA) and applying Access Control Lists to control inbound and outbound traffic.

1.2 Simulator Used

Cisco Packet Tracer was used to design, configure, and simulate the entire network.

1.3 Project Objectives

- Design a small enterprise network with an inside (trusted) and outside (untrusted) zone.
- Configure NAT and PAT on the Cisco ASA firewall to translate private IP addresses to public IP addresses.
- Configure a DNS server so that internal users can access resources using domain names.
- Allow controlled ping (ICMP) connectivity between inside users and remote servers.
- Allow inside users to browse a web page hosted on a remote (outside) web server.
- Allow outside users to access a web server hosted on the inside network.
- Block internet access for one specific inside user as a security policy demonstration.

2. Network Scenario

The network is divided into three main zones, as described below:

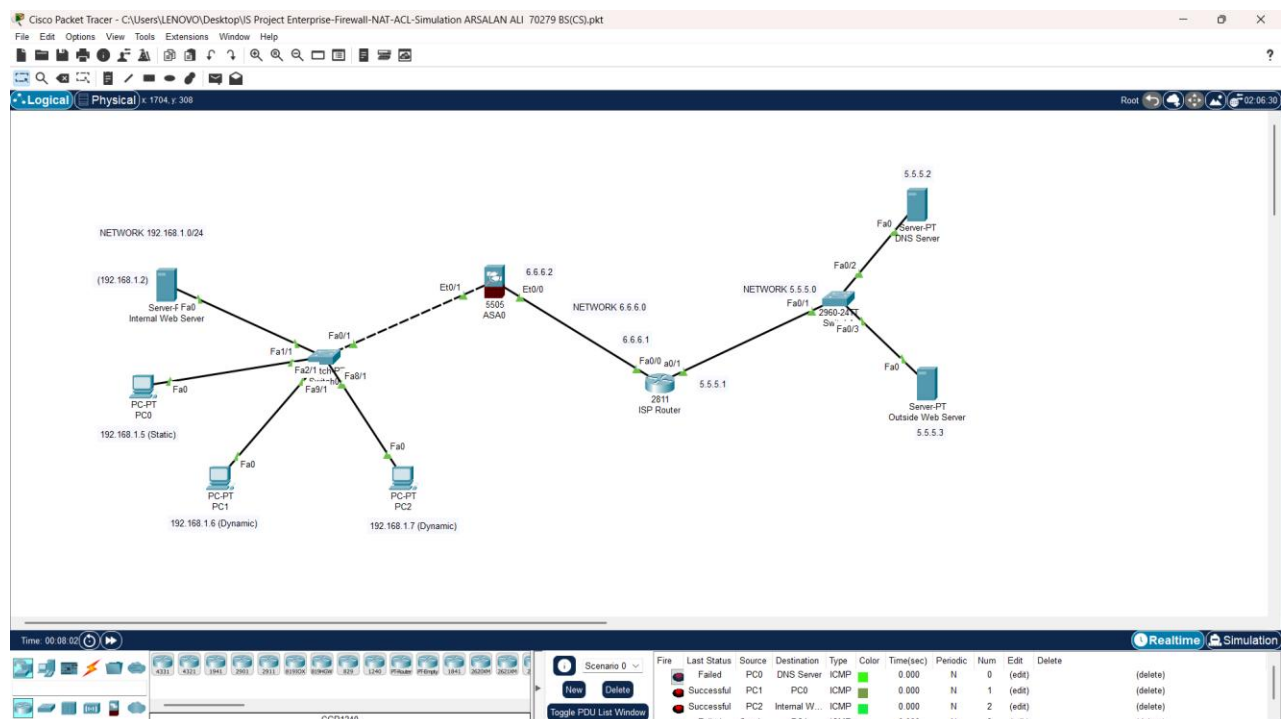
Zone	Network Address	Description
Inside Network	192.168.1.0 /24	Local company network containing PCs and an internal web server
Outside Network (to ISP)	6.6.6.0 /24	Link between the ASA firewall and the ISP router
ISP to Remote Server Network	5.5.5.0 /24	Network containing the DNS server and the outside (remote) web server

2.1 NAT Mapping (Given Scenario)

Inside Address	Translated (Public) Address	Type
192.168.1.0/24 (all inside hosts)	6.6.6.2	Dynamic PAT (Port Address Translation)
192.168.1.2 (Internal Web Server)	6.6.6.3	Static NAT

3. Network Diagram

The figure below shows the complete network topology designed in Cisco Packet Tracer, including the ASA firewall, ISP router, switches, PCs, internal web server, DNS server, and outside web server.



4. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Gateway
ASA Firewall	Inside (VLAN 1)	192.168.1.1	255.255.255.0	—
ASA Firewall	Outside (VLAN 2)	6.6.6.2	255.255.255.0	—
ISP Router	Toward ASA	6.6.6.1	255.255.255.0	—
ISP Router	Toward Switch1	5.5.5.1	255.255.255.0	—
Internal Web Server	NIC (Static)	192.168.1.2	255.255.255.0	192.168.1.1
PC0	NIC (Static)	192.168.1.5	255.255.255.0	192.168.1.1
PC1	NIC (DHCP)	192.168.1.10 (assigned)	255.255.255.0	192.168.1.1
PC2	NIC (DHCP)	192.168.1.x (assigned)	255.255.255.0	192.168.1.1
DNS Server	NIC (Static)	5.5.5.2	255.255.255.0	5.5.5.1
Outside Web Server	NIC (Static)	5.5.5.3	255.255.255.0	5.5.5.1

5. Firewall and Network Configuration

This section presents the step-by-step configuration applied on the Cisco ASA firewall and the ISP router to meet the project requirements.

5.1 ASA Interface Configuration

```

interface Vlan1
 nameif inside
 security-level 100
 ip address 192.168.1.1 255.255.255.0
!
interface Vlan2
 nameif outside
 security-level 0
 ip address 6.6.6.2 255.255.255.0
!
interface Ethernet0/0
 switchport access vlan 2
 no shutdown
!
interface Ethernet0/1
 switchport access vlan 1
 no shutdown

```

5.2 Default Route on ASA

```

route outside 0.0.0.0 0.0.0.0 6.6.6.1

```

5.3 DHCP Configuration for Inside Clients

```
dhcpd address 192.168.1.10-192.168.1.39 inside
dhcpd dns 5.5.5.2 interface inside
dhcpd enable inside
```

5.4 ISP Router Configuration

```
interface FastEthernet0/0
ip address 6.6.6.1 255.255.255.0
no shutdown
!
interface FastEthernet0/1
ip address 5.5.5.1 255.255.255.0
no shutdown
```

5.5 NAT / PAT Configuration (Requirement 1)

Static NAT was configured for the Internal Web Server so it is always reachable at a fixed public address, and dynamic PAT was configured for the rest of the inside network so multiple internal hosts can share the single outside interface address.

```
object network INTERNAL-WEBSERVER
host 192.168.1.2
nat (inside,outside) static 6.6.6.3
!
object network INSIDE-NET
subnet 192.168.1.0 255.255.255.0
nat (inside,outside) dynamic interface
```

5.6 Traffic Inspection (Supports Requirement 3)

Default inspection was enabled so that return traffic for ICMP and DNS sessions initiated from the inside network is correctly permitted back through the firewall.

```
class-map inspection_default
match default-inspection-traffic
!
policy-map global_policy
class inspection_default
inspect icmp
inspect dns
!
service-policy global_policy global
```

5.7 Access Control List — Outside to Inside (Requirement 5)

This rule allows outside users to reach the Internal Web Server (published as 6.6.6.3) over HTTP and permits ICMP so the firewall correctly passes ping replies. Two additional rules explicitly permit TCP and UDP return traffic destined to the firewall's PAT address (6.6.6.2) this was required because the simulated firewall did not automatically track return traffic for translated (PAT'd) TCP/UDP sessions, so an explicit permit rule was added for the reply traffic of inside users browsing outside websites and using DNS (Requirement 2 and 4).

```
access-list OUTSIDE-IN extended permit tcp any host 6.6.6.3 eq 80
access-list OUTSIDE-IN extended permit icmp any any
access-list OUTSIDE-IN extended permit tcp any host 6.6.6.2
access-list OUTSIDE-IN extended permit udp any host 6.6.6.2
access-group OUTSIDE-IN in interface outside
```

5.8 Access Control List — Blocking Internet for PC0 (Requirement 6)

This policy blocks all outbound traffic from host 192.168.1.5 (PC0) while allowing normal traffic for all other inside users.

```
access-list INSIDE-IN extended deny tcp host 192.168.1.5 any
access-list INSIDE-IN extended deny udp host 192.168.1.5 any
access-list INSIDE-IN extended deny icmp host 192.168.1.5 any
access-list INSIDE-IN extended permit tcp any any
access-list INSIDE-IN extended permit udp any any
access-list INSIDE-IN extended permit icmp any any
access-group INSIDE-IN in interface inside
```

5.9 DNS Server Configuration (Requirement 2)

- DNS service enabled on the DNS Server (5.5.5.2).
- A Record added: Name = webserver.local, Type = A Record, Address = 5.5.5.3
- Inside PCs configured to use 5.5.5.2 as their DNS server (via DHCP option).

5.10 Web Server Configuration (Requirements 4 and 5)

- HTTP service enabled on the Internal Web Server (192.168.1.2) with a working index.html page.
- HTTP service enabled on the Outside Web Server (5.5.5.3) with a working index.html page.

6. Testing and Results

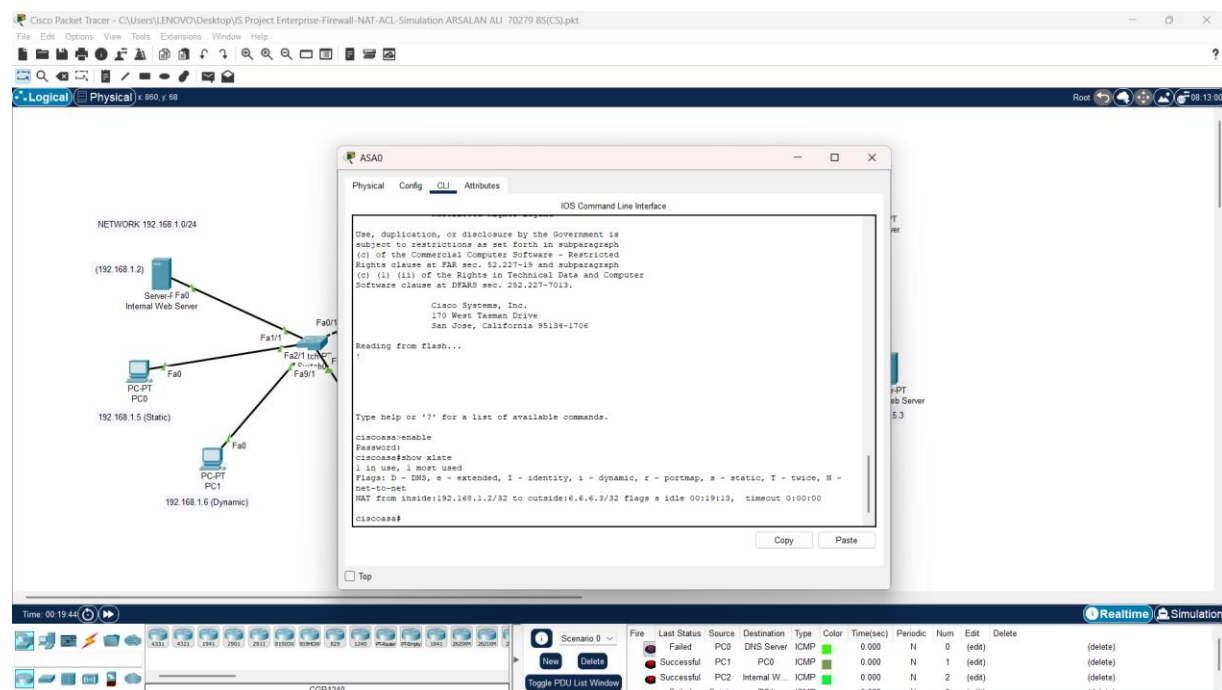
Each requirement stated in the problem statement was individually tested after configuration. The table below summarises the test performed, the command or action used, and the result observed. Screenshot evidence for every test is provided in the subsections that follow.

#	Requirement	Test Performed	Result
1	NAT / PAT configuration	show xlate on ASA — checked translation table for static and dynamic entries	Successful — 192.168.1.2 translated to 6.6.6.3 (static); other inside hosts translated to 6.6.6.2 (PAT)
2	DNS server configuration	nslookup webserver.local from inside PC	Successful — name resolved to 5.5.5.3
3	Ping from inside to remote servers	ping 5.5.5.2 and ping 5.5.5.3 from PC1 and PC2	Successful — 0% packet loss on both destinations
4	Inside users access index.html on remote web server	http://5.5.5.3 from PC1 web browser	Successful — page loaded correctly
5	Outside users access Internal Web Server	http://6.6.6.3 from DNS Server / Outside Web Server browser	Successful — Internal Web Server page loaded correctly
6	Block internet for 192.168.1.5	ping 5.5.5.2 and ping 5.5.5.3 from PC0	Successful — 100% packet loss (blocked), verified with ACL hit counters

6.1 Test Evidence — NAT / PAT Configuration (Requirement 1)

Figure 6.1 — NAT/PAT Translation Table (show xlate on ASA0)

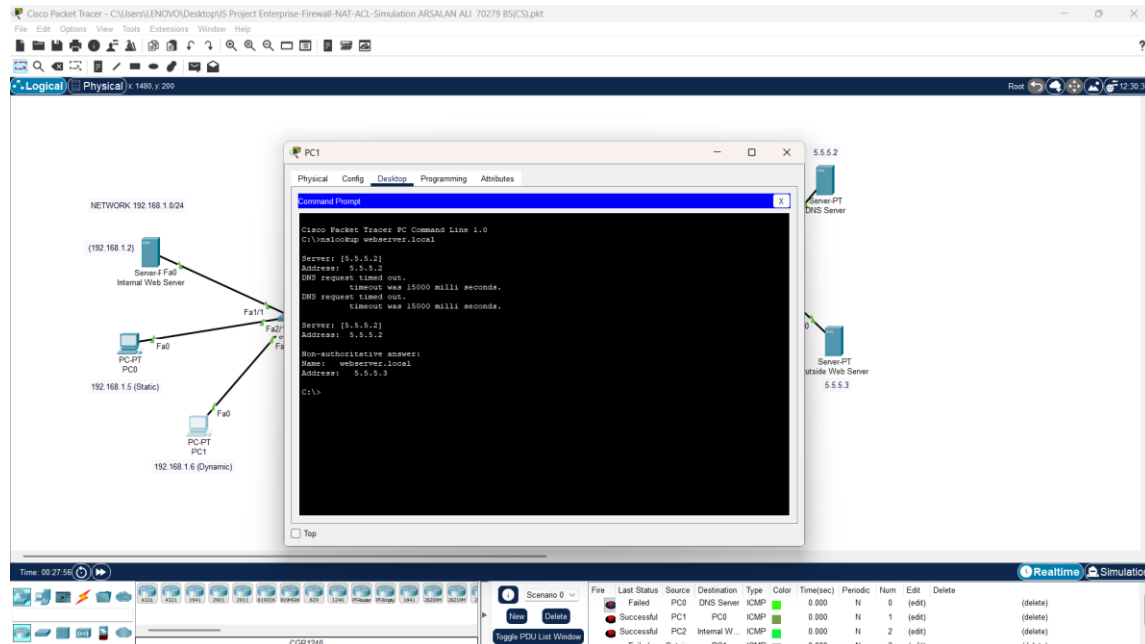
This screenshot verifies that Network Address Translation and Port Address Translation are active and functioning. The ASA's translation table confirms a static translation for the Internal Web Server (192.168.1.2 → 6.6.6.3) and dynamic PAT translations for inside hosts sharing the outside interface address (6.6.6.2).



6.2 Test Evidence — DNS Server Configuration (Requirement 2)

Figure 6.2 — DNS Name Resolution (nslookup webserver.local)

This screenshot verifies that the DNS server correctly resolves the host name webserver.local to the IP address of the Outside Web Server (5.5.5.3). The command was run from PC1's command prompt, confirming that the DNS service and A record are configured correctly.



6.3 Test Evidence — Ping Connectivity to Remote Servers (Requirement 3)

Figure 6.3 — PC1 — Ping to 5.5.5.2 and 5.5.5.3

This screenshot verifies that an inside user (PC1) can successfully reach the remote DNS Server (5.5.5.2) and Outside Web Server (5.5.5.3) using ICMP, confirming that ping traffic is permitted and correctly translated through the firewall.

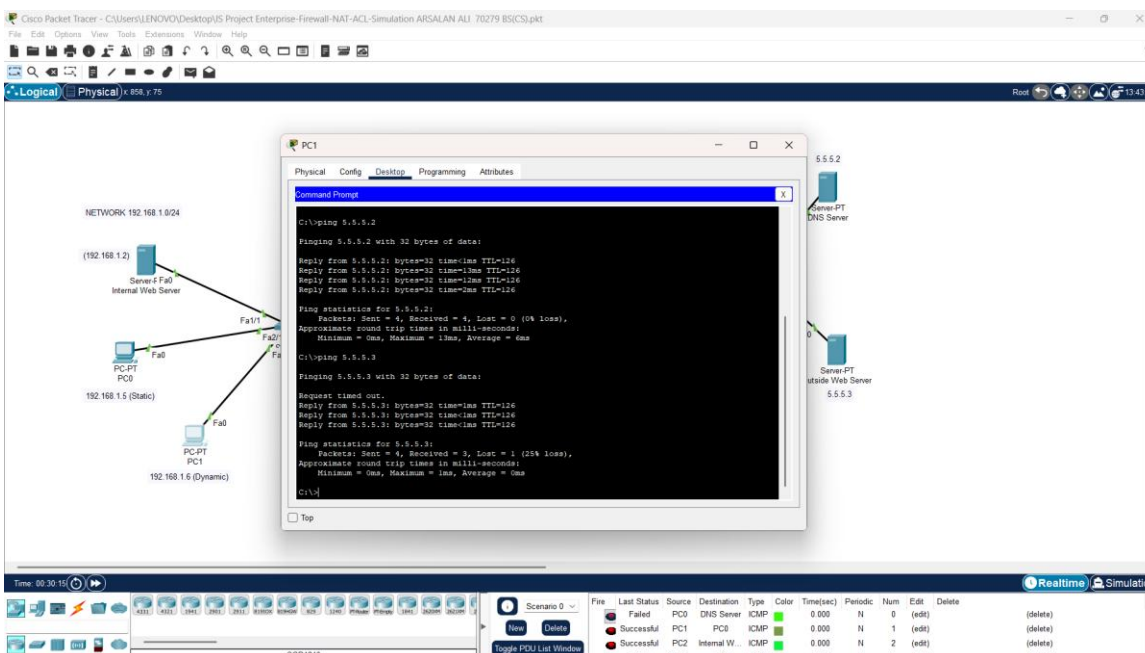
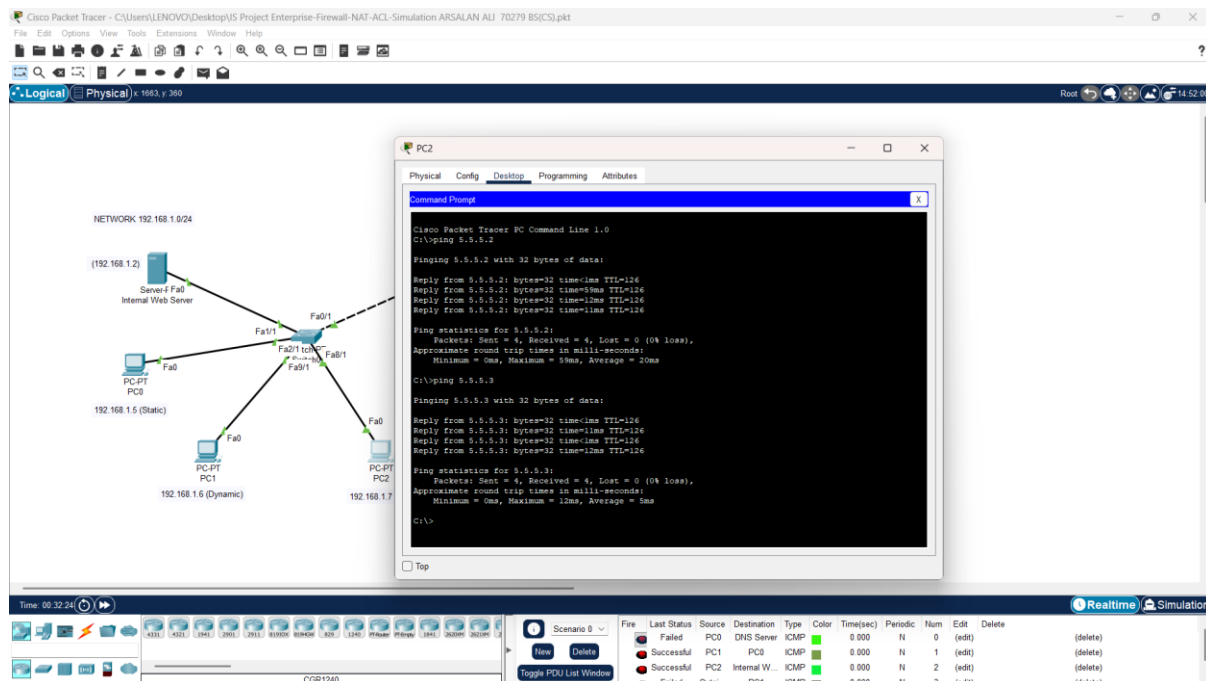


Figure 6.4 — PC2 — Ping to 5.5.5.2 and 5.5.5.3

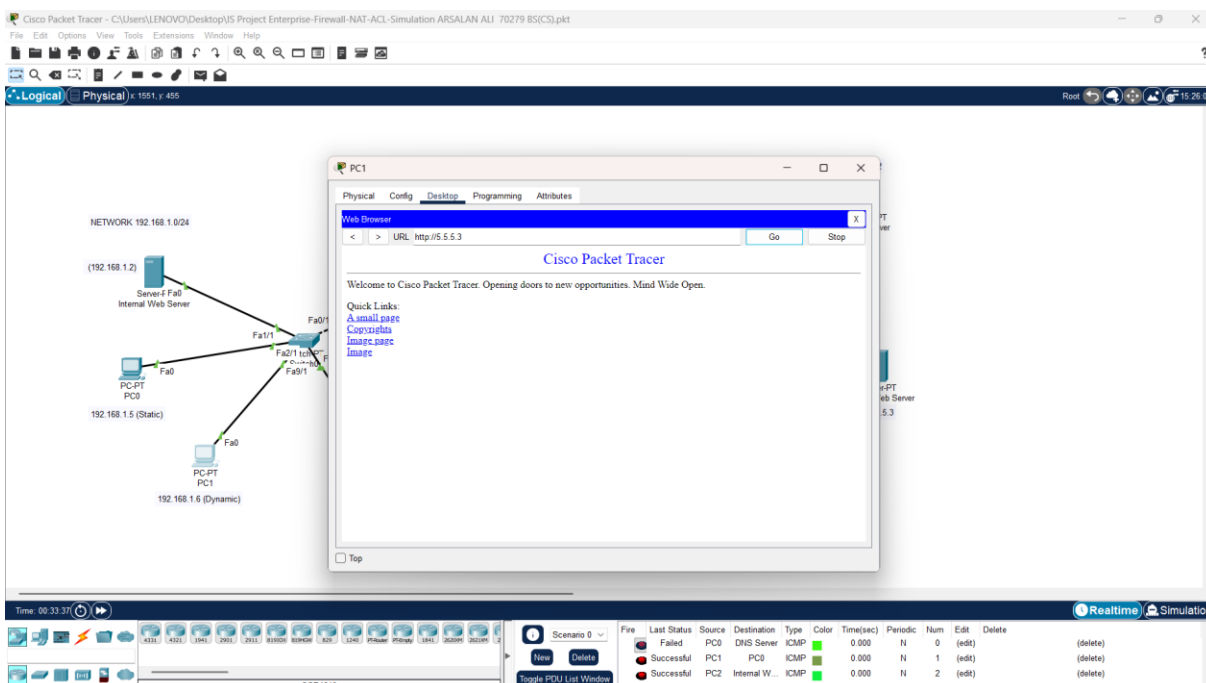
This screenshot repeats the same ping test from a second inside host (PC2), confirming that connectivity is not limited to a single user and works consistently for all permitted inside hosts.



6.4 Test Evidence — Inside Users Accessing Remote Web Server (Requirement 4)

Figure 6.5 — PC1 Web Browser — <http://5.5.5.3>

This screenshot verifies that an inside user can browse and successfully load the index.html page hosted on the Outside Web Server, confirming that outbound HTTP traffic is correctly permitted, translated, and returned through the firewall.



6.5 Test Evidence — Outside Users Accessing Internal Web Server (Requirement 5)

Figure 6.6 — DNS Server Web Browser — <http://6.6.6.3>

This screenshot verifies that an outside user (DNS Server) can reach the Internal Web Server, which is published through static NAT (192.168.1.2 → 6.6.6.3) and permitted by the OUTSIDE-IN access control list.

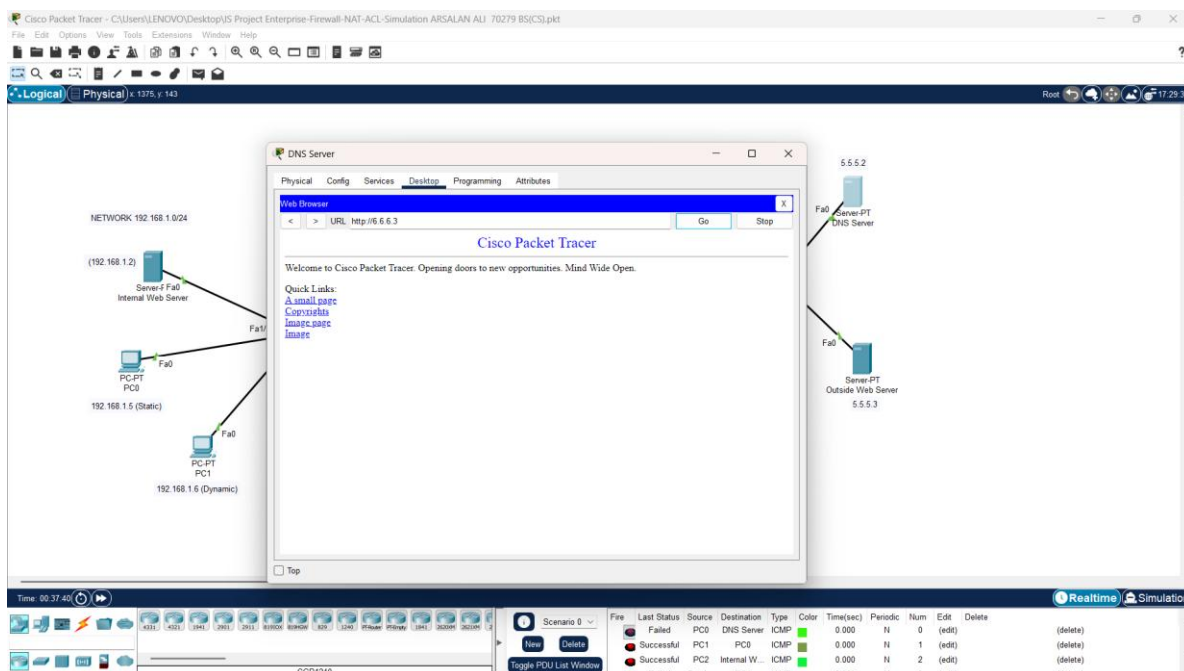
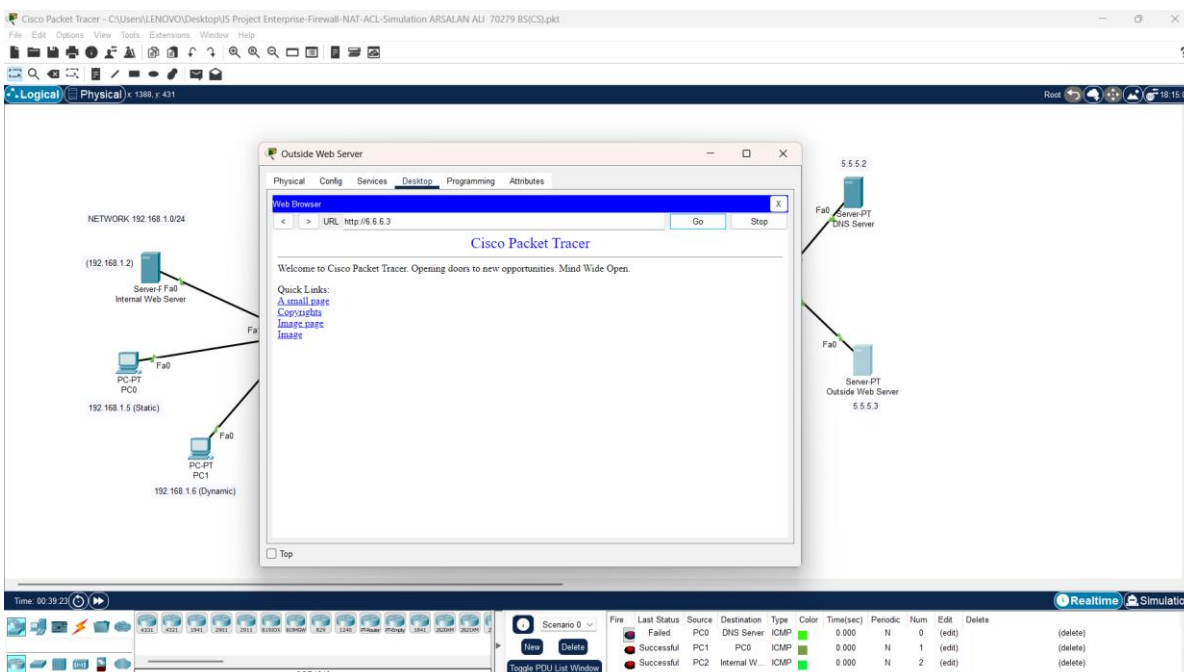


Figure 6.7 — Outside Web Server Web Browser — <http://6.6.6.3>

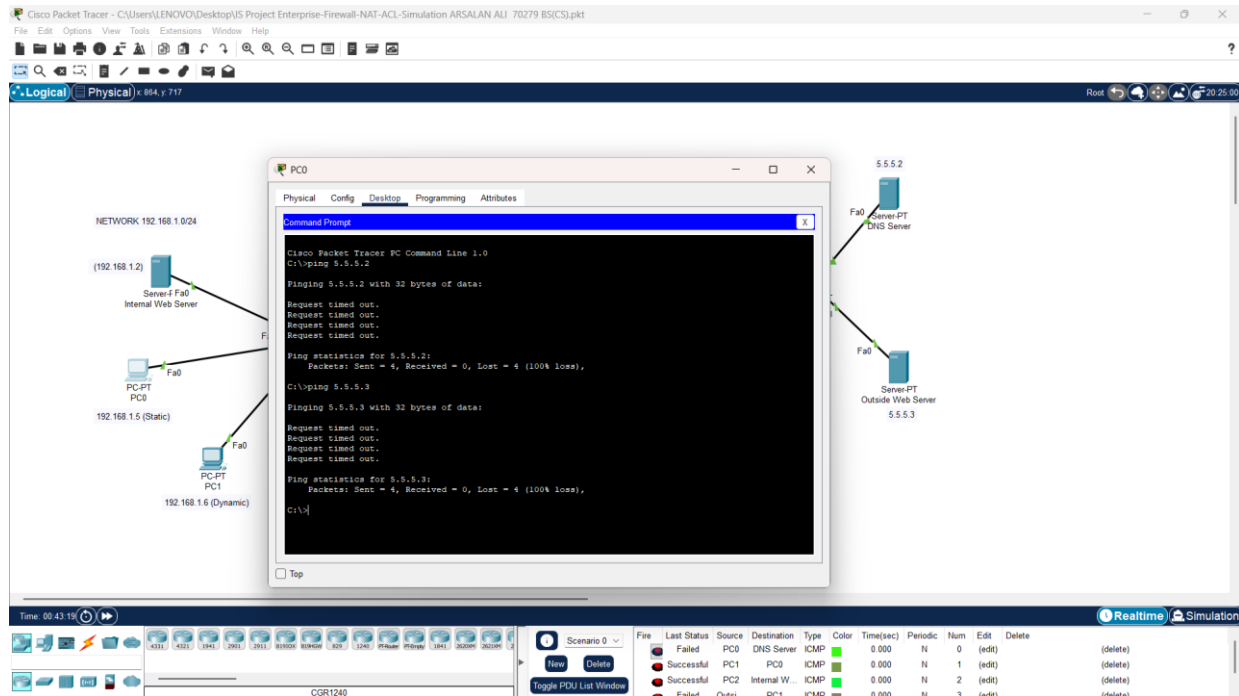
This screenshot repeats the same inbound access test from a second outside host, confirming that the Internal Web Server is reachable from the outside network in general, and not just from one specific device.



6.6 Test Evidence — Blocking Internet Access for PC0 (Requirement 6)

Figure 6.8 — PC0 — Ping to 5.5.5.2 and 5.5.5.3 (Expected: Blocked)

This screenshot verifies that the security policy blocking host 192.168.1.5 (PC0) is working correctly. All ping requests from PC0 to the remote servers time out (100% packet loss), while the same requests succeed from every other inside host.



6.7 Verification Commands Used

- show xlate — to verify active NAT/PAT translations
- show access-list — to verify ACL entries and confirm hit counters increasing during tests
- show interface ip brief — to confirm all interfaces were up and correctly addressed
- show route — to confirm the default route toward the ISP was in place

6.8 Sample Verification Output

```

ciscoasa#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list OUTSIDE-IN; 4 elements; name hash: 0x6e66eea1
access-list OUTSIDE-IN line 1 extended permit tcp any host 6.6.6.3 eq www(hitcnt=3) 0xc4b71ede
access-list OUTSIDE-IN line 2 extended permit icmp any any(hitcnt=15) 0x2b59296b
access-list OUTSIDE-IN line 3 extended permit tcp any host 6.6.6.2(hitcnt=1) 0x56ae173a
access-list OUTSIDE-IN line 4 extended permit udp any host 6.6.6.2(hitcnt=1) 0x5e4e3245
access-list INSIDE-IN; 6 elements; name hash: 0x2990bd8d
access-list INSIDE-IN line 1 extended deny tcp host 192.168.1.5 any(hitcnt=0) 0x01abbe71
access-list INSIDE-IN line 2 extended deny udp host 192.168.1.5 any(hitcnt=0) 0x7bebab86
access-list INSIDE-IN line 3 extended deny icmp host 192.168.1.5 any(hitcnt=10) 0x707a9121
access-list INSIDE-IN line 4 extended permit tcp any any(hitcnt=3) 0x8b39cc42
access-list INSIDE-IN line 5 extended permit udp any any(hitcnt=4) 0xeeba3beb
access-list INSIDE-IN line 6 extended permit icmp any any(hitcnt=17) 0x4d00ac42
  
```

7. Conclusion

This project successfully demonstrated how a Cisco ASA firewall can be used to secure an enterprise network. Network Address Translation and Port Address Translation were configured to allow inside users to reach the internet while hiding private IP addresses, and static NAT was used to publish an internal web server to the outside world. Access Control Lists were used to permit only the required traffic between the inside and outside networks, and to enforce a security policy that blocked internet access for a specific host. A DNS server was configured to allow name-based access to the remote web server.

All requirements defined in the problem statement NAT/PAT configuration, DNS setup, ping connectivity, web access in both directions, and selective access blocking were configured, tested, and verified successfully. This project provided practical, hands-on experience in enterprise firewall design, a core skill in network security administration.

8. References

- Cisco Systems — Cisco ASA 5505 Series Configuration Guide
- Cisco Packet Tracer official documentation
- Course lecture material — CMC362 Information Security Iqra University